Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S1 · Satellite types & orbits

satellite
anything that orbits something bigger
anything that orbits something bigger. The Moon is Earth's natural satellite; everything we launch is an artificial one. *A ball on a string, swung so fast the string is gravity.*

orbit
falling and missing, forever
the closed path a satellite follows around a body. It's falling toward Earth and missing, forever. *Throw a ball sideways fast enough and the ground curves away as fast as it falls.*

altitude
height above the ground, not from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

LEO / MEO / GEO · also: low / medium / geosynchronous
200–2000 km · 2000–35,786 · 35,786
Low, Medium, Geosynchronous Earth Orbit: the three height bands. LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km. *Downstairs, middle floor, penthouse.*

orbital period (T)
time for one lap: 90 min / 12 h / 24 h
Time for one revolution: $T = 2\pi\sqrt{(a^3/\mu)}$. Depends only on a and the central body.

geosynchronous (GSO)
1-day lap, any tilt
an orbit whose period is exactly one day, at any tilt. The satellite returns to the same spot in the sky once a day but may drift north-south in between.

geostationary (GEO)
1-day lap + over equator, stays put
geosynchronous AND directly over the equator with no tilt, so it appears frozen in one spot. TV dishes point at these and never move. *Sync = same time every day. Stationary = never leaves.*

equator / equatorial plane
flat sheet through Earth's middle
the flat sheet through Earth's middle, perpendicular to its spin axis. The reference plane for Earth orbits.

sidereal day
true spin vs stars, 23 h 56 m
one true rotation of Earth relative to the stars: 23 h 56 min, not 24 h. That's the real period a GEO satellite matches. *The 4-minute difference is Earth also moving around the Sun.*

astronomical satellite · also: space telescope
looks out: Hubble, JWST
a telescope in space pointed away from Earth (Hubble, JWST). *Looks out.*

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S1 · Types of interplanetary spacecraft

interplanetary
between planets, orbiting the Sun
between planets. An interplanetary spacecraft leaves Earth's neighborhood entirely, as opposed to a satellite that stays in Earth orbit.

flyby
passes once, never captured
a spacecraft that passes a planet once at high speed and keeps going. It never slows down enough to be caught. *Driving past a house and photographing it out the window.*

orbiter
brakes into orbit, stays
a spacecraft that fires its engine to slow down at the planet so gravity captures it into orbit. *Pulling into the driveway and staying.*

atmospheric probe
dives into the air, short life
a small craft that dives into a planet's air on purpose, usually on a parachute or balloon, and reports until it's crushed or runs out of power (Galileo probe at Jupiter, Huygens at Titan).

lander
touches down, stays put
a craft that survives touchdown and sits in one place doing science (Viking, InSight, Philae).

rover
lander with wheels
a lander with wheels: it can drive to new spots (Sojourner, Spirit, Opportunity, Curiosity, Perseverance).

penetrator · also: impactor
crashes in on purpose
a craft designed to slam into the surface on purpose, either to bury sensors or to kick up material for another craft to analyze (Deep Impact hit a comet; LCROSS hit the Moon).

observatory spacecraft · also: space telescope
telescope platform, target = sky
a telescope platform whose target is the sky, not a planet (Hubble, JWST, Kepler).

communications spacecraft · also: relay
forwards messages for others
a relay whose job is passing messages between other spacecraft and Earth. Mars orbiters relay for the rovers; China's Queqiao relays for the far side of the Moon.

commitment ladder · also: mission types
flyby→orbiter→probe→lander→rover→penetrator
my phrase, not the instructor's: flyby → orbiter → atmospheric → lander → rover → penetrator, ordered by how much fuel, mass, and risk each needs, and how much science it returns.

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S2 · Space vacuum conditions

ultraviolet (UV)
invisible high-energy light, sunburns
light that's a bit "bluer than blue," too high-energy for your eyes to see. It's what sunburns you. *The invisible part of sunlight that does damage.*

wavelength
size of one ripple; shorter = more energy
the size of one ripple of a light wave. Shorter wavelength = more energy = more damage. UV is shorter than visible light. *Small ripples hit harder.*

µm (micrometer) · also: micron
millionth of a meter; 0.3 µm = 300 nm
one millionth of a meter. 0.3 µm = 300 nm. A human hair is ~70 µm. *The caption said "millimeters": wrong by 1,000×.*

polymer · also: plastic
long molecule chains: Kapton, Teflon
any plastic, rubber, or similar material made of long repeating chains of molecules. Kapton, Teflon, epoxies, cable insulation. *Long molecular spaghetti.*

organic chemical bond · also: bond energy
carbon glue; UV photon beats it
the "glue" between carbon atoms that holds those chains together. Takes a certain amount of energy to break. *UV has more than that amount, so it snaps the glue.*

degradation / decomposition
material falling apart over time
the material slowly falling apart: going brittle, yellowing, cracking, flaking. *Like a garden hose left in the sun for 5 years.*

corrosion
chemical attack (rust); loose word for UV damage
chemical attack on a material, like rust on iron. The instructor used it loosely for UV damage; strictly, UV damage is bond-breaking, not corrosion. *Rust = corrosion. Sun-rotted plastic = degradation.*

UV photodegradation · also: UV degradation
photons snap polymer bonds
Breaking of chemical bonds in polymers by UV photons whose energy exceeds the bond energy (~4 eV at 300 nm vs ~3.6 eV for C–C). Also called: UV degradation

outgassing
gas leaking out of solids in vacuum
gas slowly leaking out of a solid material once it's in vacuum. *Like a new car's "new car smell": plastics releasing trapped stuff, but in space it lands on your camera lens.*

TML / CVCM
outgassing limits: < 1% / < 0.1%
Total Mass Loss and Collected Volatile Condensable Material: the two numbers a material gets from a standard

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S2 · Space vacuum conditions

radiator

panel that glows heat into space

A panel that dumps the spacecraft's waste heat into space by radiation, the only heat exit in vacuum.

MLI · also: multilayer insulation

thermal blanket · crinkly gold wrap

Stacks of thin reflective film that slow radiative heat loss. The crinkly gold/silver wrapping on spacecraft. Also called: thermal blanket

equilibrium / hot spot

settled temperature / unspread heat

Thermal equilibrium is when heat has spread out so everything sits at the same temperature. A hot spot is a place where heat has piled up and not spread yet. On Earth convection evens hot spots out; in vacuum they stay hot until conduction or radiation slowly moves the heat. *A pan on the stove: the whole pan warms evenly in air; in vacuum only the spot over the flame would glow.*

electrical discharge · also: ESD

a spark

built-up charge suddenly jumping somewhere: a spark. *The zap from a doorknob after walking on carpet.*

spacecraft charging · also: surface charging · electrostatic discharge (ESD)

static builds, no air to bleed it, arcs

the spacecraft slowly collecting electric charge from space (sunlight knocks electrons off, charged particles stick). With no air to bleed it away, it builds until it sparks. *You on a carpet, forever, with no doorknob.*

passive vs active system

no power (coating) / powered (heater)

Passive: works with no power and no decisions, by its nature (a coating, a grounding strap, a radiator, a thermal blanket). Active: does something on command, using power (a heater, a reaction wheel, a thruster, a charge-bleeding device). Engineers prefer passive where possible because it can't fail by losing power. *A window shade vs. air conditioning.*

S2 · Earth atmosphere

atmosphere

blanket of gas held by gravity

the blanket of gas held around a planet by gravity. Thick at the bottom, fading to nothing a few hundred km up. *Earth's air, all of it.*

troposphere

0–11 km, weather, colder up

bottom floor, 0–11 km. Weather, clouds, airliners. *"Tropo" = turning: the air churns here.*

stratosphere

11–50 km, ozone, warmer up

11–50 km. Calm, layered, holds the ozone layer, warms with height. *"Strato" = layered.*

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S2 · Earth atmosphere

photodissociation
light splits $O_2 \rightarrow O + O$
Light breaking a molecule apart. $O_2 + UV \rightarrow O + O$.

anomalous O · also: hot oxygen
minor extra-hot O above 300 km
a tiny extra population of unusually energetic oxygen atoms that the chart's model adds above ~300 km. Minor; not something you need to remember beyond "it's still oxygen."

ionosphere
60–1000 km, charged, bends radio, delays GPS
~60–1,000 km, overlapping the upper layers: solar UV and X-rays ionize atoms, leaving free electrons that refract and reflect radio and delay GPS signals.

geocorona
hydrogen halo glowing in UV
The faint hydrogen halo of the exosphere, glowing in UV, extending tens of thousands of km.

VLEO / LEO
below ~250 km heavy drag / up to 2000
Very Low Earth Orbit (below ~250 km, heavy drag, short life) vs Low Earth Orbit (up to 2,000 km). The green line on the chart marks the split.

reactive
eager to bond, sticky
eager to bond chemically with other stuff. *Sticky.*

oxidation
oxygen bonding to a material; rust
oxygen bonding to a material and changing it. Rust is iron oxidizing. Atomic oxygen oxidizes spacecraft surfaces; here "corrosion" is the right word. *Rust, but in orbit, and fast.*

erosion
worn away bit by bit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

erosion yield / fluence
loss per atom / total atoms that hit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

kinetic energy / relative velocity
energy of motion / closing speed
kinetic energy = energy of motion. Relative velocity = how fast two things approach each other; it doesn't matter which one is "really" moving. The s/c at 7.5 km/s hitting a still atom is the same as the atom hitting a still s/c at 7.5 km/s.

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S3 · Orbit definition & parameters

perigee / apogee · also: periapsis / apoapsis
closest, fastest / farthest, slowest
closest and farthest points of an orbit around Earth, measured from Earth's center. *"Peri" = near, "apo" = far, "gee" = Earth.*

perihelion / aphelion
same, around the Sun
The closest and farthest points of an orbit around the Sun. Same idea as perigee and apogee, with the suffix changed for the body: -gee for Earth, -helion for the Sun, -lune for the Moon, -apsis for anything. Earth's own perihelion is in early January, about 147 million km from the Sun; aphelion in early July, about 152 million km. *Perigee and apogee, Sun edition.*

apsis (apsides)
either extreme point
either of those two extreme points. The line joining them is the long axis of the ellipse.

line of apsides · also: major axis
perigee↔apogee line
The straight line through periapsis and apoapsis, passing through the central body. Its half-length is a. Also called: major axis

inclination (i) · also: tilt
tilt vs equator; 0° flat, 90° polar
the tilt angle between the orbit's plane and the reference plane (the equator, for Earth). 0° flat, 90° over the poles.

ecliptic
plane of Earth's orbit; reference for Sun orbits
the flat sheet Earth's orbit around the Sun lies in. The reference plane for orbits around the Sun. *Not the same as the equator; they're tilted 23.4° apart.*

plane of reference
what you measure tilt from
the flat sheet you measure tilt from. Equator for Earth orbits, ecliptic for Sun orbits.

vernal equinox (♈) · also: First Point of Aries
reference direction · fixed zero direction in space
a fixed "12 o'clock" direction in space to measure angles from: the direction to the Sun on the first day of spring. The ♈ symbol (Aries) marks it on diagrams. *North on a compass, but for space.*

ascending node
hoop crosses equator going north
the two points where a tilted orbit pierces the reference plane. The ascending node is the one where the satellite is crossing from south to north. *Where the hoop pokes up through the table.*

line of nodes
where orbital plane cuts equator

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S3 · Earth orbits & constellations

rotation period
Earth spins once: 23 h 56 m
how long one full trip around takes. LEO ~90 min, GPS 12 h, GEO 24 h. Higher = longer. *Bigger circle, slower runner.*

altitude vs distance from center
GEO 35,786 alt = 42,164 from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

coverage / global coverage
what a sat sees / everything, always
Coverage is the patch of Earth a satellite can see or talk to at a given moment (its footprint). Global coverage means every point on Earth is covered all the time, which one satellite can never do and a constellation can. *A flashlight's circle of light vs. a stadium's floodlights.*

resolution
how much detail; closer = sharper
How much detail an instrument can make out: the smallest thing it can tell apart. Depends on the optics and on distance, so a LEO camera can see a car while a GEO weather camera sees whole storm systems. Higher orbit = bigger coverage but lower resolution. *Reading a street sign from 10 feet vs. from a mile.*

orbital plane (constellation) · also: Walker constellation
several rings so sats aren't bunched
The flat sheet the orbit lies in. Always passes through the center of the body being orbited. Why GEO must be over the equator.

revisit time
how long until same spot again
How long until a satellite (or constellation) sees the same spot again. Shorter with more satellites or higher altitude.

GPS / GLONASS / Galileo / BeiDou
US / Russia / EU / China nav constellations
Global Positioning System, the US navigation constellation at ~20,200 km in MEO.

Iridium
66 sats, 780 km, sat phones anywhere
A company and its satellite constellation: 66 active satellites at about 780 km in six near-polar orbital planes, giving satellite-phone and data coverage everywhere on Earth including the poles and oceans. Named for element 77 because the original plan had 77 satellites. *The swarm that makes a sat phone work in the middle of the ocean.*

data relay / transfer time
sat → ground; higher = more delay
Data relay is passing information from one spacecraft to another or down to Earth, often through a dedicated relay satellite. Transfer time is how long the signal takes; radio travels at light speed, so it grows with distance: ~0.12 s one way to GEO, 4–24 minutes to Mars. Higher orbits mean longer delay. *A game of telephone across the solar system, with*

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S3 · Transfers & flybys

time of flight (TOF) · also: transfer time
Hohmann = half the ellipse's period
how long the trip takes. For a Hohmann it's half the ellipse's period. *Your Section 1 quiz question.*

launch window / synodic period
lineup repeats; Mars every 26 months
the stretch of days when the planets are lined up right for a Hohmann. Earth–Mars: about every 26 months. *Repeat from S1 primer.*

phase angle · also: lead angle
how far ahead target must be: Mars ~44°
how far ahead the target planet must be when you launch so it's there when you arrive. Mars: ~44° ahead. Not on the slide; it's why the window exists.

impulsive burn · also: impulsive maneuver
engine firing treated as instant
An engine firing treated as instantaneous. Hohmann assumes two of them. Also called: impulsive maneuver

circularization burn · also: apogee kick
2nd burn rounds the ellipse
The second burn of a Hohmann: at apoapsis, speed up to round the ellipse into the target circle. Also called: apogee kick

plane change · also: inclination change
tilting the orbit; very expensive
A burn to tilt the orbital plane. Very expensive: $\Delta v = 2v \cdot \sin(\Delta i/2)$. Done at apogee where v is small. Also called: inclination change

bi-elliptic transfer
3-burn alternative, only for huge jumps
A three-burn alternative that goes far out and comes back. Beats Hohmann only for very large radius ratios. Named so you recognize it.

gravity field
region where the planet's pull matters
the region around a planet where its pull matters. It's what bends the flyby path.

relative velocity
speed as seen from Mars vs from Sun
your speed as seen from something else (Mars, say) rather than from the Sun. The flyby doesn't change your speed relative to Mars, only relative to the Sun. *Repeat.*

vector / magnitude / direction
arrow / its length / which way
an arrow: a quantity with both a size and a direction. Velocity is a vector. The slide's arrow pictures are vector addition: spacecraft-relative-to-Mars plus Mars-relative-to-Sun equals spacecraft-relative-to-Sun. *Two pushes added tip to tail.*